

02

FRIDAY

January

002-363
1ST WEEK '26

NOVEMBER 2025

Sun	30	2	9	16	23
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DECEMBER 2025

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Homework 9

08.00

1) Given

09.00

$$P(y_i=1) = p_i = \frac{e^{x_i\beta}}{1+e^{x_i\beta}}, \quad y_i \in \{0,1\}, \quad i=1, \dots, n$$

10.00

11.00

This is a logistic regression model with one parameter β .

12

i) To find, Log-likelihood function of $L(\beta)$

13.00

For Bernoulli, $P(y_i) = p_i^{y_i} (1-p_i)^{1-y_i}$

14.00

Substitute p_i from given,

15.00

$$P(y_i) = \left(\frac{e^{x_i\beta}}{1+e^{x_i\beta}} \right)^{y_i} \left(\frac{1}{1+e^{x_i\beta}} \right)^{1-y_i}$$

16.00

17.00

$$= \frac{e^{x_i\beta y_i}}{(1+e^{x_i\beta})^{y_i} (1+e^{x_i\beta})^{1-y_i}}$$

18.00

$$\Rightarrow P(y_i) = \frac{e^{x_i\beta y_i}}{(1+e^{x_i\beta})}$$

Notes

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03

January

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1ST WEEK

SATURDAY

Likelihood:
$$L(\beta) = \prod_{i=1}^n \frac{e^{x_i \beta y_i}}{1 + e^{x_i \beta}}$$

Log-likelihood:
$$\log L(\beta) = \sum_{i=1}^n [x_i \beta y_i - \log(1 + e^{x_i \beta})]$$

ii) To find First & Second Derivatives.

First derivative:
$$\frac{dL}{d\beta} = \frac{d}{d\beta} \log L(\beta)$$

$$= \frac{d}{d\beta} \sum_{i=1}^n [x_i \beta y_i - \log(1 + e^{x_i \beta})]$$

$$= \sum_{i=1}^n \left[x_i y_i - \frac{x_i e^{x_i \beta}}{1 + e^{x_i \beta}} \right]$$

We know that
$$p_i = \frac{e^{x_i \beta}}{1 + e^{x_i \beta}}$$

$$\Rightarrow \frac{dL}{d\beta} = \sum_{i=1}^n x_i (y_i - p_i)$$

Second derivative:
$$\frac{d^2 L}{d\beta^2} = \sum_{i=1}^n \left[-x_i^2 \frac{e^{x_i \beta}}{(1 + e^{x_i \beta})^2} \right]$$

04 Sunday

05

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MONDAY

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Using $p_i(1-p_i)$

$$\Rightarrow \frac{d^2 L}{d\beta^2} = - \sum_{i=1}^n x_i^2 p_i(1-p_i)$$

iii) Newton Raphson update:

$$\beta^{(t+1)} = \beta^{(t)} - \frac{L'(\beta)}{L''(\beta)}$$

We know that,

$$L'(\beta) = \sum_{i=1}^n x_i(y_i - p_i)$$

$$L''(\beta) = - \sum_{i=1}^n x_i^2 p_i(1-p_i)$$

$$\Rightarrow \beta^{(t+1)} = \beta^{(t)} + \frac{\sum_{i=1}^n x_i(y_i - p_i)}{\sum_{i=1}^n x_i^2 p_i(1-p_i)}$$

Notes

where

$$p_i = \frac{e^{x_i \beta(t)}}{1 + e^{x_i \beta(t)}}$$

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January

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06

TUESDAY

iv) To show,

08.00

$$E(\hat{\beta}) = \beta, \quad \text{Var}(\hat{\beta}) = \frac{1}{\sum_{i=1}^n x_i^2 p_i^* (1-p_i^*)}$$

09.00

For maximum likelihood estimators.

10.00

11.00

$$\hat{\beta} \approx N\left(\beta, \frac{1}{I(\beta)}\right) \text{ where } I(\beta) \text{ is the Fisher information}$$

12.00

Recall log likelihood; $l(\beta) = \sum_{i=1}^n [x_i \beta y_i - \log(1 + e^{x_i \beta})]$

13.00

By Fisher Information definition

14.00

$$I(\beta) = -E[l''(\beta)]$$

15.00

We know ~~the~~ $l''(\beta)$ from previous derivations. Substituting, we get,

16.00

$$I(\beta) = -E\left[-\sum_{i=1}^n x_i^2 p_i^* (1-p_i^*)\right]$$

17.00

18.00

$$I(\beta) = \sum_{i=1}^n x_i^2 p_i^* (1-p_i^*)$$

Notes

p_i depends on x_i, β , not on randomness.

So, expectation doesn't change it.

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WEDNESDAY

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Now, $\text{Var}(\hat{\beta}) = \frac{1}{I(\beta)}$

$\Rightarrow \text{Var}(\hat{\beta}) = \frac{1}{\sum_{i=1}^n x_i^2 p_i (1-p_i)}$

Since β is unknown, we replace with MLE $\hat{\beta}$

$p_i = \frac{e^{x_i \hat{\beta}}}{1 + e^{x_i \hat{\beta}}}$

$\Rightarrow \text{Var}(\hat{\beta}) = \frac{1}{\sum_{i=1}^n x_i^2 \hat{p}_i (1-\hat{p}_i)}$

From MLE theory, $\hat{\beta} \xrightarrow{\text{asymptotically}} N(\beta, \dots)$

$\Rightarrow E(\hat{\beta}) \approx \beta$

————— X —————

v) Hypothesis test: $H_0: \beta = 0$

Notes

We use a Wald test.

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08

January

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2ND WEEK

THURSDAY

Test Statistic: $Z = \frac{\hat{\beta}}{\sqrt{\text{Var}(\hat{\beta})}}$

$$\Rightarrow Z = \hat{\beta} \cdot \sqrt{\sum_{i=1}^n x_i^2 \hat{p}_i (1 - \hat{p}_i)}$$

Decision Rule, Under H_0 , $Z \sim N(0, 1)$

Reject if: $|Z| > z_{\alpha/2}$

If rejected $\rightarrow x$ significantly affects y .

If not $\rightarrow$ no significant effect.

15.00

2) We now take a Bayesian approach:

Likelihood $\rightarrow$ Same as problem 1.

Prior $\rightarrow \beta \sim N(0, \sigma^2)$

i) Posterior Density (up to proportionality)

From Problem 1, $L(\beta) = \prod_{i=1}^n \frac{e^{x_i \beta y_i}}{1 + e^{x_i \beta}}$

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08.00 Normal prior, $\pi(\beta) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{\beta^2}{2\sigma^2}\right)$

09.00

Apply Bayes' Theorem,

10.00

$$\pi(\beta | \text{data}) \propto L(\beta) \pi(\beta)$$

11.00

Multiply Likelihood & Prior.

12

$$\pi(\beta | \text{data}) \propto \left[\prod_{i=1}^n \frac{e^{x_i \beta y_i}}{1 + e^{x_i \beta}} \right] \cdot \exp\left(-\frac{\beta^2}{2\sigma^2}\right)$$

13.00

14.00

Combine Exponent Terms,

15.00

$$\prod e^{x_i \beta y_i} = e^{\sum x_i \beta y_i}$$

16.00

17.00

$$\Rightarrow \pi(\beta | \text{data}) \propto \exp\left(\sum_{i=1}^n x_i \beta y_i\right) \cdot \exp\left(-\frac{\beta^2}{2\sigma^2}\right) \cdot \prod_{i=1}^n \frac{1}{1 + e^{x_i \beta}}$$

18.00

$$\Rightarrow \pi(\beta | \text{data}) \propto \exp\left(\sum_{i=1}^n x_i \beta y_i - \frac{\beta^2}{2\sigma^2}\right) \prod_{i=1}^n \frac{1}{1 + e^{x_i \beta}}$$

Notes

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2ND WEEK

SATURDAY

ii) Show Log Posterior is Concave.

08.00

09.00

$$\log \pi(\beta | \text{data}) = \sum_{i=1}^n x_i \beta y_i - \sum_{i=1}^n \log(1 + e^{x_i \beta}) - \frac{\beta^2}{2\sigma^2} + C$$

10.00

First Derivative,

11.00

$$\frac{d}{d\beta} = \sum_{i=1}^n x_i (y_i - p_i) - \frac{\beta}{\sigma^2}$$

12.00

Second Derivative,

13.00

$$\frac{d^2}{d\beta^2} = - \sum_{i=1}^n x_i^2 p_i (1 - p_i) - \frac{1}{\sigma^2}$$

14.00

$$\Rightarrow p_i(1 - p_i) > 0$$

15.00

$$\Rightarrow x_i^2 > 0$$

16.00

$$\Rightarrow \frac{1}{\sigma^2} > 0$$

17.00

$$\text{So, } \frac{d^2}{d\beta^2} < 0$$

11 Sunday

18.00

$\Rightarrow$ Log posterior is concave //

Notes

_____ X _____

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MONDAY

January

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3RD WEEK '26

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iii) Improper prior $\pi(\beta) \propto 1$

08.00

Posterior becomes, $\pi(\beta | \text{data}) \propto L(\beta)$

09.00

$\Rightarrow$ Posterior $\propto$ Likelihood

10.00

$\Rightarrow$ Maximizing posterior $\equiv$ Maximizing Likelihood

11.00

12

Therefore, the relationship between

13.00

posterior & maximum likelihood estimator is that

14.00

$$\boxed{\text{Posterior mode} = \text{MLE } \hat{\beta}}$$

15.00

————— X —————

16.00

iv) Log Bayes Factor:

17.00

We compare: $H_0: \beta = 0$

18.00

$H_1: \beta = 1$

Notes

By Bayes factor definition,

$$B = \frac{L(\beta=1)}{L(\beta=0)}$$

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3RD WEEK

13

TUESDAY

08.00 Likelihood, $L(\beta) = \prod \frac{e^{x_i \beta y_i}}{1 + e^{x_i \beta}}$

09.00 Take Log, $\log B = \log L(1) - \log L(0)$

10.00 Evaluate Each Term,

11.00 For $\beta=1$: $\log L(1) = \sum x_i y_i - \sum \log(1 + e^{x_i})$

12 For $\beta=0$: $\log L(0) = \sum 0 - \sum \log(2) = -n \log 2$

13.00 $\Rightarrow \log B = \sum x_i y_i - \sum \log(1 + e^{x_i}) + n \log 2$

14.00 Rearranged form:

15.00 $\log B = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \log(1 + e^{x_i})$

16.00

17.00 ✓) Approximate Distribution for $\log B$

18.00 We know that,

Notes $\log B = \sum_{i=1}^n [x_i y_i - \log(1 + e^{x_i})]$

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January

WEDNESDAY

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Under $H_0: \beta = 0$

08.00

$$\Rightarrow p_i = \frac{1}{2}, y_i \sim \text{Bernoulli}(1/2)$$

09.00

10.00

Mean, $E[x_i y_i] = x_i \cdot \frac{1}{2}$

11.00

$$\Rightarrow E[\log B] = \sum \left[\frac{x_i}{2} - \log(1 + e^{x_i}) \right]$$

12

Variance, $\text{Var}(x_i y_i) = x_i^2 \cdot \frac{1}{4}$

13.00

14.00

$$\Rightarrow \text{Var}(\log B) = \sum \frac{x_i^2}{4}$$

15.00

Apply CLT, since sum of independent terms,

16.00

$$\log B \approx N \left(\sum \left[\frac{x_i}{2} - \log(1 + e^{x_i}) \right], \sum \frac{x_i^2}{4} \right)$$

17.00

18.00

Notes